

RESEARCH PROPOSAL

Title: Trustworthy GenAI for Radiology Reports via Conformal Validation of Statements

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1. Abstract

Generative AI is rapidly changing the technical landscape of medical image interpretation. In radiology, large vision-language models can already produce fluent, plausible reports from imaging studies. Yet fluency is not clinical validity. A generated report may contain statements that are unsupported by the image, omit clinically important observations, or express certainty when the visual evidence is ambiguous. Such errors may hinder their incorporation into routine clinical practice, as they raise unresolved concerns regarding reliability, uncertainty, and safety.

This proposal develops a framework for evidence-constrained generative radiology. It moves beyond standard report-generation evaluation, which often relies on text similarity, label extraction, or uncalibrated factuality scoring. Instead, it probes the statistical validity of image-conditioned clinical statements implicitly asserted by the text, addressing their dependencies while preserving clinically relevant modifiers. Structured conformal predictions will be used to obtain calibrated prediction for sets of related radiology statements with formal guarantees. These guarantees will then be translated into explicit generative behaviors, namely assertion, qualification, or abstention, thereby aligning model behavior with clinical intent. The research will investigate the mathematical properties of conformalized statements, the emergence of epistemic and aleatoric uncertainty through conformal set behavior, and the limits of safe generation under imbalanced or rare findings, partial observability and domain shift. The outcome is a theoretically grounded and practically applicable framework for medical GenAI systems in which free-text outputs must be constrained by evidence.

2. Scientific background and state of the art

Radiology GenAI and the safety gap. Radiology reports are evidential clinical documents: they communicate what is justified by imaging data. Thus, the central risk in report generation is not stylistic failure, but clinically consequential statements that are unsupported, insufficiently qualified, or omitted despite being relevant.

Current evaluation methods for radiology report generation do not fully address this safety problem. Lexical metrics such as BLEU or ROUGE are poorly aligned with clinical correctness. Label-based scores provide useful summaries but collapse rich report language into a limited set of findings. Entity- and relation-based approaches, such as RadGraph, move closer to clinical structure by extracting findings, anatomical locations, and relations (Jain et al., 2021). Newer evaluation frameworks such as GREEN attempt to identify clinically significant errors in generated reports (Ostmeier et al., 2024). Studies of report-generation progress and metrics have also shown that clinically meaningful metrics such as RadGraph F1 and RadCliQ provide more useful evaluation than text overlap alone (F. Yu et al., 2023). These developments demonstrate that the field is moving away from surface-level text similarity. However, they still largely evaluate generated reports after the fact, without providing a principled statistical mechanism that determines what a model may safely assert, qualify, or withhold.

From report generation to evidence-constrained communication. Recent radiology vision-language models

and report-generation systems, including large multimodal and dialog-oriented models, demonstrate the rapid progress of the field (Pellegrini et al., 2023). Methods for improving factual completeness and consistency have emphasized that generated reports must be evaluated not only for incorrect statements but also for missing clinically important information (Miura et al., 2021). The proposed project builds on this direction but changes the target of analysis: from the generated report as a whole to the set of dependent image-conditioned clinical statements it communicates.

Structured report information and report-derived labels. Report-derived labels and information extraction are essential for large-scale medical AI. CheXpert introduced uncertainty-aware labels for chest radiographs (Irvin et al., 2019), while CheXbert showed that BERT-based labelers can improve automatic radiology report labeling using expert annotations and rule-based supervision (Smit et al., 2020). MIMIC-CXR and MIMIC-CXR-JPG provide large public resources of chest radiographs, free-text reports, and structured labels that have enabled much of the modern literature in chest imaging AI (Johnson, Pollard, Berkowitz, et al., 2019; Johnson, Pollard, Greenbaum, et al., 2019). These datasets and labelers are important empirical resources, but the proposal does not reduce the scientific contribution to a specific dataset or label list. They will be used to instantiate and test a general framework for evidence-calibrated statements.

Conformal prediction and medical uncertainty. Uncertainty quantification is central to medical AI. A medical GenAI system should not only produce an answer; it should know when the evidence is insufficient. Conformal prediction provides a distribution-free framework for constructing prediction sets with finite-sample coverage guarantees under exchangeability (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos* and Bates, 2023). In its standard form, conformal prediction produces a set that contains the true output with probability at least $1 - \alpha$. This is attractive for medicine because uncertainty can be communicated explicitly rather than hidden behind a single prediction. Related work has developed conformalized quantile regression (Romano et al., 2019), risk-controlling prediction sets (Bates et al., 2021), conformal risk control (Angelopoulos et al., 2024), and multi-label prediction sets with limited false positives (Fisch* et al., 2022). These methods provide important statistical tools, but they do not directly define the clinical unit of validation for free-text medical generation.

Conditional calibration, class imbalance, and rare statements. Marginal coverage over all statements may be misleading in radiology. Statement distributions are highly imbalanced: common normal or chronic findings dominate the data, while rare but clinically important findings may have limited calibration support. A system can appear well calibrated overall while being unreliable for rare findings, uncertain statements, specific anatomical regions, or high-risk clinical concepts. Class-conditional and clustered conformal prediction directly address the difficulty of obtaining stronger coverage guarantees when classes are numerous or imbalanced (Ding et al., 2023). Work on conditional histograms, label-noise robustness, missing values, and corrupted labels further demonstrates that calibration must be adapted to the structure and imperfections of real data (Sesia and Romano, 2021; Einbinder et al., 2022; Zaffran et al., 2023; Feldman* et al., 2025). This project aims to bring these ideas to image-conditioned clinical statements, where the conditioning variables are not only labels but also semantic, clinical, and representation-based neighborhoods.

Medical imaging uncertainty beyond scalar prediction. Medical imaging introduces additional structure that is often absent from standard classification problems. Predictions may be spatial, multi-label, correlated across anatomy, affected by acquisition protocol, and tied to clinical language rather than to a single endpoint. Recent work on uncertainty quantification for imaging has therefore emphasized the need to move beyond independent

pixel-wise or label-wise confidence scores toward structured uncertainty objects, including spatially coherent uncertainty, conformal masks, and clinically interpretable uncertainty displays (Belhasin et al., 2024; Kutiel* et al., 2023). These ideas are highly relevant here: the proposed framework treats uncertainty as a property of image-conditioned clinical communication. The object to be calibrated is not only an image classifier output, but the evidence relationship between an image and a statement.

The remaining gap. The key scientific gap is not simply classifier calibration, but the calibration of the relationship between an image and a dependent set of image-conditioned clinical statements. Report statements may encode presence, absence, uncertainty, severity, laterality, chronicity, temporal change, or omission, and these states may imply, qualify, contradict, or constrain one another. The proposed project reframes report generation as evidence-constrained clinical communication, centered on a validation unit that preserves clinical meaning while enabling structured, conditional, and multidimensional calibration.

3. Research objectives and specific aims

The overall objective is to develop a framework for evidence-constrained generative radiology, in which generated reports are validated as sets of image-conditioned clinical statements rather than as free text or isolated labels. The central hypothesis is that report generation becomes safer and more clinically useful when each statement is assigned a calibrated evidence status relative to the image, and report language is constrained accordingly.

The main methodological contribution is structured conformal validation of dependent clinical statements. The project will define the validation unit, represent dependencies among statements, construct image-conditioned nonconformity scores, calibrate admissible evidence assignments, and translate the calibrated sets into report-level behaviors such as assertion, qualification, abstention, or flagging for review.

Aim 1: Define image-conditioned clinical statements as validation units. We will formalize each report statement as a structured object encoding a clinical concept, assertion state, and modifiers such as location, laterality, severity, temporal change, or uncertainty. This aim establishes the basic unit on which statistical validation is performed.

Aim 2: Model dependencies among statements and define structured nonconformity. We will represent relations among normalized statements using a corpus-level dependency graph and combine image-statement evidence with statement-compatibility terms. The resulting structured nonconformity score will measure whether a joint assignment of evidence states is supported by both the image and the learned statement dependencies.

Aim 3: Develop structured conformal calibration for dependent statements. We will apply conformal calibration to structured nonconformity scores to obtain sets of admissible joint evidence assignments. Unlike standard conformal prediction for isolated labels, the calibrated object will describe which combinations of statement evidence states are statistically supported.

Aim 4: Address imbalance, rare findings, and multidimensional clinical risk. We will evaluate calibration across clinically meaningful strata, including evidence state, anatomical region, statement type, and finding-frequency group. Rare findings will be grouped or handled conservatively when calibration data are insufficient. We will also use scalar clinically weighted losses, where appropriate, to study communication risks such as unsupported assertion, contradiction, omission, overconfidence, and inappropriate uncertainty.

Aim 5: Extend validation from stated content to omissions and report behavior. We will probe candidate

statements absent from the report to detect possible omissions of clinically important findings. Calibrated evidence will then be translated into report-level behavior: assert when supported, qualify when ambiguous, abstain or flag when unsupported, and indicate possible omissions when absent statements are image-supported.

4. Detailed description of the proposed research

4.1 Validation unit: image-conditioned clinical statements

Let $X \in \mathcal{X}$ denote the observed radiological imaging study, such as a radiograph, CT volume, or MRI scan, and let T denote a generated or human-written report. The framework does not assume access to the internal report generator. It treats the report as a set of structured clinical statements $S(T) = s_1, \dots, s_m$. Each statement $s_j = (c_j, a_j, \mu_j) \in \mathcal{S}$ encodes a clinical concept c_j , an assertion state a_j , and optional clinically relevant modifiers μ_j , such as location, laterality, severity, temporal change, or uncertainty language.

For each statement, the validation task is to determine its evidence status with respect to the image, $Y(X, s_j) \in \mathcal{Y}$, where, in the simplest setting, $\mathcal{Y} = \{\text{supported}, \text{contradicted}, \text{underdetermined}\}$. Thus, the basic unit of validation is not a word, sentence, or dataset label alone, but an image-conditioned clinical statement.

4.2 Representing dependencies among statements

To account for dependencies among clinical statements, we define a corpus-level graph over normalized statements extracted from radiology reports. Graph-based representations have been used for pathology co-occurrence and structured radiology information Chen et al., 2020; Li et al., 2019; Jain et al., 2021. Here, the graph serves as a dependency prior within a structured conformal validation framework, rather than as a standalone graph-learning model.

Let $G_{\mathcal{D}} = (V_{\mathcal{D}}, E_{\mathcal{D}})$ denote a corpus-level dependency graph constructed from a corpus $\mathcal{D}$ of D reports. The nodes $V_{\mathcal{D}} = \{V_1, \dots, V_K\}$ are normalized clinical statements, and $E_{\mathcal{D}} = (e_{ij}) \in [-1, 1]^{K \times K}$ is a weighted adjacency matrix encoding pairwise relations among statements. Each node V_k is associated with a report-occurrence vector $(v_{k,1}, \dots, v_{k,D})$, $v_{k,d} \in \{-1, 0, 1\}$, where $v_{k,d} = 1$ indicates assertion in report d , $v_{k,d} = -1$ indicates explicit absence or negation, and $v_{k,d} = 0$ indicates non-mention or unresolved status.

The edge weight e_{ij} represents the relation between statements V_i and V_j . Positive values indicate compatibility or co-occurrence, negative values indicate contradiction or clinical inconsistency, and values near zero indicate weak or no dependency. Edge weights may be estimated from report-level co-occurrence, negation and uncertainty patterns, and RadGraph-style entity and relation annotations Jain et al., 2021.

For an individual report T , the relevant dependency structure is the subgraph induced by the statements in $S(T)$, possibly augmented by candidate high-priority statements for omission detection. These dependencies are incorporated into the structured nonconformity score through compatibility terms between evidence states. Thus, the graph provides a practical representation of statement dependencies, while the main contribution is conformal calibration of dependent image-conditioned statements.

4.3 Image-statement evidence and structured nonconformity

Image conditioning is case-specific and complementary to corpus-level dependency learning. For each statement s_j , an image-statement evidence model produces $\phi_j(y; X, s_j)$, $y \in \mathcal{Y}$, or an equivalent evidence score measuring how well the image supports, contradicts, or leaves undetermined the statement.

The dependency model provides pairwise compatibility factors $\psi_{jk}(y_j, y_k; R_{jk})$, which evaluate whether the joint evidence states of two related statements are consistent with their learned relation. The structured evidence for a joint assignment Y_T is modeled as a product of image-conditioned statement evidence and corpus-level dependency compatibility:

$$Q(X, T, Y_T) = \prod_{j=1}^m \phi_j(Y_j; X, s_j) \prod_{j < k} \psi_{jk}(Y_j, Y_k; R_{jk}). \quad (1)$$

Here, ϕ_j measures how well the image supports the evidence state of statement s_j , while ψ_{jk} measures whether the joint evidence states of two statements are consistent with their learned relation. The structured nonconformity score is then defined as the negative log of this evidence, with an additional omission term:

$$A_{\text{str}}(X, T, Y_T) = - \sum_{j=1}^m \log \phi_j(Y_j; X, s_j) - \lambda \sum_{j < k} \log \psi_{jk}(Y_j, Y_k; R_{jk}) + \mu A_{\text{omit}}(X, T). \quad (2)$$

The score A_{str} is not optimized directly as a training loss. The image-evidence model and dependency model are learned, and the structured score defines the nonconformity measure used for conformal calibration.

4.4 Structured conformal calibration

Given a calibration set of image-report pairs with reference evidence states, conformal calibration will be applied to the structured nonconformity scores. For a target level $1 - \alpha$, the calibrated prediction object is a set of admissible joint evidence assignments,

$$\Gamma_\alpha(X, T) = Y_T : A_{\text{str}}(X, T, Y_T) \leq q_\alpha,$$

where q_α is the calibration quantile. This object provides calibrated evidence for the report as a dependent set of statements, rather than independent confidence scores for isolated findings.

The resulting evidence sets will be translated into report behavior. A uniquely supported statement may be asserted. A statement whose evidence set includes underdetermination should be qualified or deferred. A contradicted statement should be withheld or flagged. Broad or unstable evidence sets indicate that the report should be reviewed rather than presented as definitive.

4.5 Stratified calibration and multidimensional risk control

We will evaluate the structured conformal object across pre-defined clinically meaningful strata. Let

$$A_i = A_{\text{str}}(X_i, T_i, Y_{T_i})$$

denote the structured nonconformity score for calibration case i , and let

$$C_i = C(X_i, T_i, Y_{T_i}) \in \mathcal{C}$$

denote a stratum, such as statement type, anatomical region, evidence state, or finding-frequency group. The evidence state may distinguish present, absent, uncertain, and non-mentioned statements, allowing class imbalance to be evaluated explicitly.

When a stratum c contains sufficient calibration samples, we will estimate a stratum-specific conformal threshold

$$q_{\alpha, c} = \text{Quantile}_{1-\alpha} \{A_i : C_i = c\},$$

and define

$$\Gamma_{\alpha}^{\text{strat}}(X, T) = \{Y_T : A_{\text{str}}(X, T, Y_T) \leq q_{\alpha, C(X, T, Y_T)}\}.$$

We will address two practical challenges in calibration: label noise and limited calibration data. Since report-derived calibration labels may be noisy or uncertain, we will assess sensitivity to label noise and, when needed, consider noise-aware conformal calibration following Penso et al. Penso* et al., 2025. When calibration data are limited, as expected for rare findings, rare labels will be handled conservatively by grouping anatomically or clinically related labels before calibration, consistent with clustered class-conditional conformal prediction Ding et al., 2023, or by using the global threshold while reporting empirical coverage, set size, and abstention rate separately.

The multidimensional component concerns the type of clinical communication error. For each test case, we define a risk vector

$$R_i(\lambda) = (R_i^{\text{assert}}(\lambda), R_i^{\text{contr}}(\lambda), R_i^{\text{omit}}(\lambda), R_i^{\text{over}}(\lambda), R_i^{\text{unc}}(\lambda)),$$

corresponding to unsupported assertion, contradiction, omission, overconfident wording, and inappropriate uncertainty. The parameter λ controls the conservativeness of report behavior, for example the transition from assertion to qualification or abstention.

To summarize clinically weighted risk, we will define a bounded scalar loss

$$L_i(\lambda) = \sum_{\ell} w_{\ell} R_i^{\ell}(\lambda), \quad \sum_{\ell} w_{\ell} = 1,$$

where larger weights may be assigned to clinically severe errors such as omission of critical findings or unsupported assertion of severe pathology. When $L_i(\lambda)$ is monotone in λ , conformal risk control will be used to select λ so that

$$\mathbb{E}[L(\lambda)] \leq \alpha_{\text{risk}}$$

under the standard exchangeability assumption Angelopoulos et al., 2024. The individual components of R_i will also be reported separately, so that the safety–informativeness tradeoff is not hidden by a single weighted score.

4.6 Candidate-statement probing for omissions

Validating only statements that appear in a report cannot detect clinically important omissions. We will therefore construct a candidate set $\mathcal{Q}(X, T)$ of statements that are absent from the report but potentially relevant to the image. Candidates may be obtained from the clinical statement vocabulary, high image-evidence scores, retrieval in the learned representation space, or clinician-defined high-priority findings.

For each candidate $q \in \mathcal{Q}(X, T)$, we evaluate an augmented statement set

$$S^+(T, q) = S(T) \cup \{q\}.$$

The conformal machinery is then applied to the possible evidence states of q , yielding a calibrated set

$$\Gamma_\alpha^q(X, T) = \{y_q : \exists Y_T \text{ such that } A_{\text{str}}(X, T, S^+(T, q), (Y_T, y_q)) \leq q_\alpha\}.$$

If $\Gamma_\alpha^q(X, T)$ strongly supports a present or clinically reportable state for a statement absent from the report, the system flags a possible omission. If the evidence is ambiguous, the omission is qualified rather than asserted. If the candidate is not supported, no omission is flagged. These omission indicators contribute to the report-level risk component R^{omit} defined above. Together, Sections 4.1–4.6 define the proposed validation pipeline, from structured statement extraction and dependency representation to calibrated evidence assignments and omission probing. The next sections describe the datasets and preliminary experiments used to instantiate and evaluate this framework.

4.7 Datasets and clinical statement spaces

The project will use two complementary public radiology domains. For chest radiography, we will use MIMIC-CXR and MIMIC-CXR-JPG Johnson, Pollard, Berkowitz, et al., 2019; Johnson, Pollard, Greenbaum, et al., 2019, which provide chest radiographs paired with free-text reports and structured report-derived labels. This domain supports experiments on common findings such as atelectasis, cardiomegaly, consolidation, edema, pleural effusion, pneumonia, and pneumothorax.

For MRI, we will use MR-RATE Forthmus, 2026, which provides brain and spine MRI studies paired with radiology reports and pathology labels. This domain will test the framework under longer reports, heterogeneous anatomy, multi-sequence imaging, and more complex statement structures.

In both domains, the clinical statement space will be constructed from normalized report findings and modifiers, including anatomical location, laterality, severity, uncertainty, and negation. Frequent findings will support stratum-specific calibration, while rare but clinically important findings will be grouped or evaluated conservatively without claiming label-specific validity.

4.8 Evaluation, clinical input, and contingency plan

Evaluation will include coverage, conformal set size, abstention rate, subgroup behavior, and communication-risk components: unsupported assertion, contradiction, omission, overconfident wording, and inappropriate uncertainty. Ablations will compare independent finding-level calibration, structured calibration without dependency terms,

and the full dependency-aware framework.

Clinical input will be obtained through collaboration with radiology experts at Soroka Medical Center. This collaboration will support definition of clinically meaningful statement vocabularies, prioritization of high-risk omissions, interpretation of uncertainty language, and qualitative review of selected failure cases. The core experiments will rely on public datasets; any use of local clinical data will be pursued only subject to the required institutional approvals.

If statement extraction is noisy, we will restrict the vocabulary to high-confidence normalized statements. If dependency estimation is unstable, we will use simpler co-occurrence statistics, clinical rules, or expert-defined relations. If rare findings lack sufficient calibration samples, they will be grouped by clinical or anatomical similarity, or evaluated using global calibration without label-specific validity claims. If full report-generation integration is not feasible, the framework will be evaluated as a post-hoc validator of generated or reference reports.

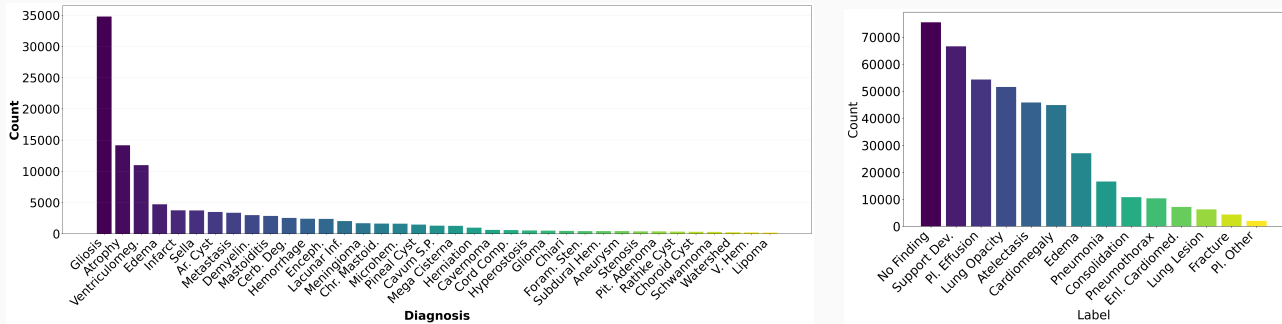


Figure 1: Label-frequency histograms for the two dataset domains. Left: MR-RATE brain/spine MRI labels. Right: MIMIC-CXR/CheXpert-style chest radiography labels. Labels were abbreviated for readability. The histograms illustrate long-tailed candidate statement vocabularies and motivate calibration strategies for frequent, imbalanced, and clinically high-priority findings.

4.9 Experimental Setup and Results

Table 1: Preliminary finding-level conformal prediction results. Pos/Neg coverage denotes empirical conformal coverage on definite positive and negative cases. Decisive accuracy is computed only for singleton conformal sets. Unc.→Abs. denotes expert-uncertain cases assigned the non-singleton set {Present, Absent}.

Pathology	Baseline			Conformal prediction			
	TPR	TNR	Acc.	Pos Cov.	Neg Cov.	Decisive Acc.	Unc.→Abs.
Atelectasis	28.0%	90.9%	80.6%	91.4%	88.3%	76.8%	56.8%
Cardiomegaly	56.2%	91.6%	86.4%	90.9%	89.3%	84.3%	33.7%
Consolidation	57.7%	73.3%	70.3%	93.5%	93.0%	79.2%	63.4%
Pleural Effusion	80.7%	85.0%	83.6%	88.9%	87.8%	86.7%	20.1%
Pneumothorax	44.9%	91.0%	84.4%	94.7%	90.2%	79.0%	54.5%
Mean	53.5%	86.4%	81.1%	91.9%	89.7%	81.2%	45.7%

As a preliminary feasibility study, we evaluated a simplified finding-level conformal prediction framework on 23,462 chest radiographs from 14,876 CheXpert patients across five pathologies. Unlike the proposed framework, this experiment uses independent binary finding labels and does not yet model report text, modifiers, omissions, or

dependencies among structured clinical statements. Ground-truth labels extracted from radiology reports indicate whether each finding is *present*, *absent*, or *uncertain*. To prevent patient-level leakage, the data were partitioned into disjoint training (67%), validation (10%), calibration (10%), and test (13%) sets, with label distributions preserved across splits.

The baseline model was a DenseNet121 multi-label classifier (Huang et al., 2017), trained with binary cross-entropy loss. To define a controlled binary setting, uncertain cases were excluded from training and calibration, which used only definite positive and definite negative labels. We then applied class-conditional conformal prediction, computing separate nonconformity thresholds for present and absent labels to reduce the effect of class imbalance, with target coverage $1 - \alpha = 0.9$.

At test time, uncertain cases were retained to probe whether expert uncertainty corresponds to conformal ambiguity. Singleton prediction sets, {Present} or {Absent}, were treated as decisive predictions, while the non-singleton set {Present, Absent} was treated as abstention. As shown in Table 1, conformal prediction achieved close to nominal coverage for definite positive and negative labels. The results also show that expert uncertainty and conformal abstention are related but not equivalent: uncertainty-to-abstention varies substantially across pathologies. This supports the need for the proposed statement-level framework, where uncertainty, omission, modifiers, and dependencies are modeled explicitly rather than treated as a third class.

Milestone number	1	Due date	30/9/2027
Milestone title	Structured clinical-statement representation and dependency prior		
Description	Define image-conditioned clinical statements and modifiers; construct the normalized statement vocabulary; and represent dependencies among statements using a corpus-level graph. Corresponds to Sections 4.1–4.2.		

Milestone number	2	Due date	30/9/2028
Milestone title	Structured nonconformity and conformal calibration		
Description	Develop image-statement evidence scores, combine them with dependency compatibility terms, and calibrate admissible joint evidence assignments using structured conformal prediction. Includes stratified calibration, imbalance, rare findings, and multidimensional risk. Corresponds to Sections 4.3–4.5.		

Milestone number	3	Due date	30/9/2029
Milestone title	Omission probing, report behavior, and full empirical evaluation		
Description	Extend validation to candidate statements absent from the report, translate calibrated evidence into assertion, qualification, abstention, or omission flags, and complete empirical evaluation on the proposed datasets. Corresponds to Sections 4.6–4.9.		

5. Significance, innovation and potential benefits of the proposed research

This project reframes the safety problem of generative medical AI from report similarity to evidence-constrained communication: what is a model clinically justified in saying, given the image?

The main innovation is a structured validation unit: the image-conditioned clinical statement. Unlike labels or whole-report scores, statements preserve clinically relevant meaning, including presence, absence, uncertainty, modifiers, contradictions, and omissions. The second innovation is structured conformal validation of dependent statements. Rather than calibrating isolated findings, the framework calibrates admissible joint evidence assignments, accounting for image support and statement dependencies. The third innovation is the translation of calibrated evidence into report behavior: assertion, qualification, abstention, or omission flagging.

The expected benefit is a safety layer for medical GenAI systems that goes beyond fluency and factuality scoring. It can support clinician-supervised drafting, pre-deployment auditing, model comparison, and monitoring of unsupported assertions, uncertainty mismatch, contradictions, and omissions. More broadly, the project provides a framework for free-text medical AI outputs that must remain accountable to observed evidence.

6. Applicability

The immediate application is AI-assisted radiology reporting. A generated report can be parsed into clinical statements, and the proposed validator can identify which statements are image-supported, uncertain, contradicted, or possibly missing. The radiologist remains responsible for the final report, but receives a structured safety signal.

A second application is pre-deployment auditing of generative radiology systems. The framework can be used to measure statement-level coverage, abstention, set size, rare-finding behavior, uncertainty mismatch, contradiction rates, and omission patterns before clinical use.

A third application is post-deployment monitoring. Changes in calibrated set size, abstention rate, or subgroup performance may indicate reduced reliability and motivate review or recalibration.

Although the empirical focus is radiology, the framework is applicable to other medical domains in which generated text makes data-grounded clinical statements.

7. Work plan and Gantt

#	Milestones	6	12	18	24	30	36
1	Statement representation and vocabulary (Sec. 4.1)	■	■				
2	Dependency representation among statements (Sec. 4.2)	■	■	■			
3	Structured evidence and nonconformity score (Sec. 4.3)		■	■	■		
4	Structured conformal calibration and risk control (Secs. 4.4–4.5)			■	■	■	
5	Candidate-statement probing and report behavior (Secs. 4.6–4.7)				■	■	■
6	Dataset experiments and empirical evaluation (Secs. 4.8–4.9)		■	■	■	■	■

References

- Angelopoulos, A. N., S. Bates, A. Fisch, L. Lei, and T. Schuster (2024). “Conformal Risk Control”. In: *International Conference on Learning Representations*.
- Angelopoulos*, A. N. and S. Bates (2023). “Conformal Prediction: A Gentle Introduction”. In: *Foundations and Trends in Machine Learning* 16.4, pp. 494–591.
- Bates, S., A. N. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan (2021). “Distribution-Free, Risk-Controlling Prediction Sets”. In: *Journal of the ACM* 68.6, pp. 1–34.
- Belhasin, O., Y. Romano, D. Freedman, E. Rivlin, and M. Elad (2024). “Principal Uncertainty Quantification with Spatial Correlation for Image Restoration Problems”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.5, pp. 3321–3333.
- Chen, B., J. Li, G. Lu, H. Yu, and D. Zhang (2020). “Label Co-Occurrence Learning With Graph Convolutional Networks for Multi-Label Chest X-Ray Image Classification”. In: *IEEE Journal of Biomedical and Health Informatics* 24.8, pp. 2292–2302.
- Ding, T., A. N. Angelopoulos, S. Bates, M. I. Jordan, and R. J. Tibshirani (2023). “Class-Conditional Conformal Prediction with Many Classes”. In: *Advances in Neural Information Processing Systems*. Vol. 36, pp. 64555–64576.
- Einbinder, B.-S., S. Feldman, S. Bates, A. N. Angelopoulos, A. Gendler, and Y. Romano (2022). “Label Noise Robustness of Conformal Prediction”. In: *arXiv preprint arXiv:2209.14295*.
- Feldman*, S., S. Bates, and Y. Romano (2025). “Conformal Prediction with Corrupted Labels: Uncertain Imputation and Robust Re-weighting”. In: *arXiv preprint arXiv:2505.04733*.
- Fisch*, A., T. Schuster, T. Jaakkola, and R. Barzilay (2022). “Conformal Prediction Sets with Limited False Positives”. In: *International Conference on Machine Learning*, pp. 6514–6532.
- Forithmus (2026). *MR-RATE: Brain and Spine MRI Volumes Paired with Radiology Reports, Metadata, and Pathology Labels*. Hugging Face Dataset.
- Huang, G., Z. Liu, L. Van Der Maaten, and K. Q. Weinberger (2017). “Densely Connected Convolutional Networks”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4700–4708.
- Irvin, J., P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghgoo, R. Ball, K. Shpanskaya, et al. (2019). “CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison”. In: *AAAI Conference on Artificial Intelligence*.
- Jain, S., A. Agrawal, A. Saporta, S. Q. H. Truong, D. N. Duong, T. Bui, P. Chambon, Y. Zhang, M. P. Lungren, A. Y. Ng, C. P. Langlotz, and P. Rajpurkar (2021). “RadGraph: Extracting Clinical Entities and Relations from Radiology Reports”. In: *arXiv preprint arXiv:2106.14463*.
- Johnson, A. E. W., T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C.-y. Deng, R. G. Mark, and S. Horng (2019). “MIMIC-CXR: A De-identified Publicly Available Database of Chest Radiographs with Free-text Reports”. In: *Scientific Data* 6.317.
- Johnson, A. E. W., T. J. Pollard, N. R. Greenbaum, M. P. Lungren, C.-y. Deng, Y. Peng, Z. Lu, R. G. Mark, S. J. Berkowitz, and S. Horng (2019). “MIMIC-CXR-JPG: A Large Publicly Available Database of Labeled Chest Radiographs”. In: *arXiv preprint arXiv:1901.07042*.
- Kutiel*, G., R. Cohen, M. Elad, D. Freedman, and E. Rivlin (2023). “Conformal Prediction Masks: Visualizing Uncertainty in Medical Imaging”. In: *Machine Learning for Health*, pp. 163–176.

- Li, C. Y., X. Liang, Z. Hu, and E. P. Xing (2019). “Knowledge-Driven Encode, Retrieve, Paraphrase for Medical Image Report Generation”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 33. 01, pp. 6666–6673.
- Miura, Y., Y. Zhang, E. B. Tsai, C. P. Langlotz, and D. Jurafsky (2021). “Improving Factual Completeness and Consistency of Image-to-Text Radiology Report Generation”. In: *arXiv preprint arXiv:2010.10042*.
- Ostmeier, S., J. Xu, Z. Chen, M. Varma, L. Blankemeier, C. Bluethgen, A. E. Michalson, M. Moseley, C. Langlotz, A. S. Chaudhari, and J.-B. Delbrouck (2024). “GREEN: Generative Radiology Report Evaluation and Error Notation”. In: *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 374–390.
- Pellegrini, C., E. Ozsoy, B. Busam, N. Navab, and M. Keicher (2023). “RaDialog: A Large Vision-Language Model for Radiology Report Generation and Conversational Assistance”. In: *arXiv preprint arXiv:2311.18681*.
- Penso*, C., J. Goldberger, and E. Fetaya (2025). “Conformal Prediction of Classifiers with Many Classes based on Noisy Labels”. In: *Proceedings of the Fourteenth Symposium on Conformal and Probabilistic Prediction with Applications*. Vol. 266. Proceedings of Machine Learning Research. PMLR, pp. 82–95.
- Romano, Y., E. Patterson, and E. J. Candès (2019). “Conformalized Quantile Regression”. In: *Advances in Neural Information Processing Systems*. Vol. 32.
- Sesia, M. and Y. Romano (2021). “Conformal Prediction using Conditional Histograms”. In: *arXiv preprint arXiv:2105.08747*.
- Shafer, G. and V. Vovk (2008). “A Tutorial on Conformal Prediction”. In: *Journal of Machine Learning Research* 9, pp. 371–421.
- Smit, A., S. Jain, P. Rajpurkar, A. Pareek, A. Y. Ng, and M. P. Lungren (2020). “CheXbert: Combining Automatic Labelers and Expert Annotations for Accurate Radiology Report Labeling Using BERT”. In: *Conference on Empirical Methods in Natural Language Processing*, pp. 1500–1519.
- Vovk, V., A. Gammerman, and G. Shafer (2005). *Algorithmic Learning in a Random World*. Springer.
- Yu, F., M. Endo, R. Krishnan, I. Pan, A. Tsai, E. P. Reis, E. K. U. N. Fonseca, H. M. H. Lee, Z. S. H. Abad, A. Y. Ng, and P. Rajpurkar (2023). “Evaluating Progress in Automatic Chest X-ray Radiology Report Generation”. In: *Patterns* 4.9, p. 100802.
- Zaffran, M., A. Dieuleveut, J. Josse, and Y. Romano (2023). “Conformal Prediction with Missing Values”. In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, pp. 40578–40604.